

NICKOLAS SIMONS

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EDUCATION

University of Utah

Master of Entertainment Arts and Engineering, Engineering

Salt Lake City, UT

Aug. 2026 – May 2028

University of Illinois at Urbana-Champaign

Bachelor of Science, Computer Science

Minor in Game Studies and Design

Urbana, IL

Aug. 2022 – May 2025

GPA: 3.96

EXPERIENCE

University of Utah, Division of Games

Teaching Assistant - Survey of Games, Intro to Game Design

Salt Lake City, UT

Aug. 2026 – Dec. 2026

University of Illinois, stu/dio

Programming Lead

Urbana, IL

Aug. 2024 – Dec. 2025

- Managed a team of 12 programmers across 8 projects, conducting code reviews and providing planning feedback to ensure code quality and maintainability
- Maintained Perforce infrastructure and integrated Horde for automated builds and Helix Swarm for code review
- Established coding standards and development workflows to improve code maintainability across projects

Gameplay Programmer

April 2024 – Dec. 2025

- Designed and implemented gameplay features and tools for VR platforms using C++ and Blueprints, with an emphasis on accessibility, performance, and usability in classroom environments
- Optimized gameplay code and materials to meet VR performance targets using Unreal Insights and Nvidia Nsight

PROJECTS

Master Dancer VR | UE5, C++, Blueprint

May 2024 – Dec 2025

- Built three quantized movement-based rhythm games end-to-end, developing core gameplay architecture using C++ and Blueprint
- Created graphical level authoring tools to streamline designer workflows and support rapid iteration
- Implemented a system for anchoring and rendering subtitles in 3D space for VR applications, with support for localization

VRchaecology | UE5, C++, Blueprint

May 2025 – Nov 2025

- Designed and implemented a modular quest system with editor tooling using C++ to guide user actions through reusable scripting logic and contextual prompts
- Developed an interactive, tool-based VR excavation system using C++, simulating surface digging and stratigraphic layer analysis
- Implemented shared sublevel instances in UE5 to maintain continuity across multiple uses of a shared environment, ensuring consistent gameplay and scene behavior, and simplifying the workflow for content teams

Multiplayer Action Game | UE5, C++, Blueprint, ASIO, C#, GraphQL, MongoDB

Nov. 2025 – Present

- Built a multiplayer combat framework using C++ and Epic's Gameplay Ability System, implementing client-side prediction and correction for movement abilities
- Developed a utility service using C#, MongoDB, and GraphQL to integrate with matchmaking, allowing queries for game state information prior to joining a session
- Created a system for generating replicated procedural environments using Epic's new Procedural Content Generation framework

TECHNICAL SKILLS

Proficiencies: Gameplay Engineering, DevOps, Networking, Engine Development

Languages: C++, C, C#, SQL, Python, Haskell, UE Blueprint

Tools: Unreal Engine 5, Perforce, Git, Visual Studio, Unity 6, Docker, Horde, Jenkins

Technologies: SDL3, OpenGL, Dear ImGui, CMake, ASIO (C++), Hot Chocolate (GraphQL), Linux